

Blue Apron Training

HAAS CNCs Overview

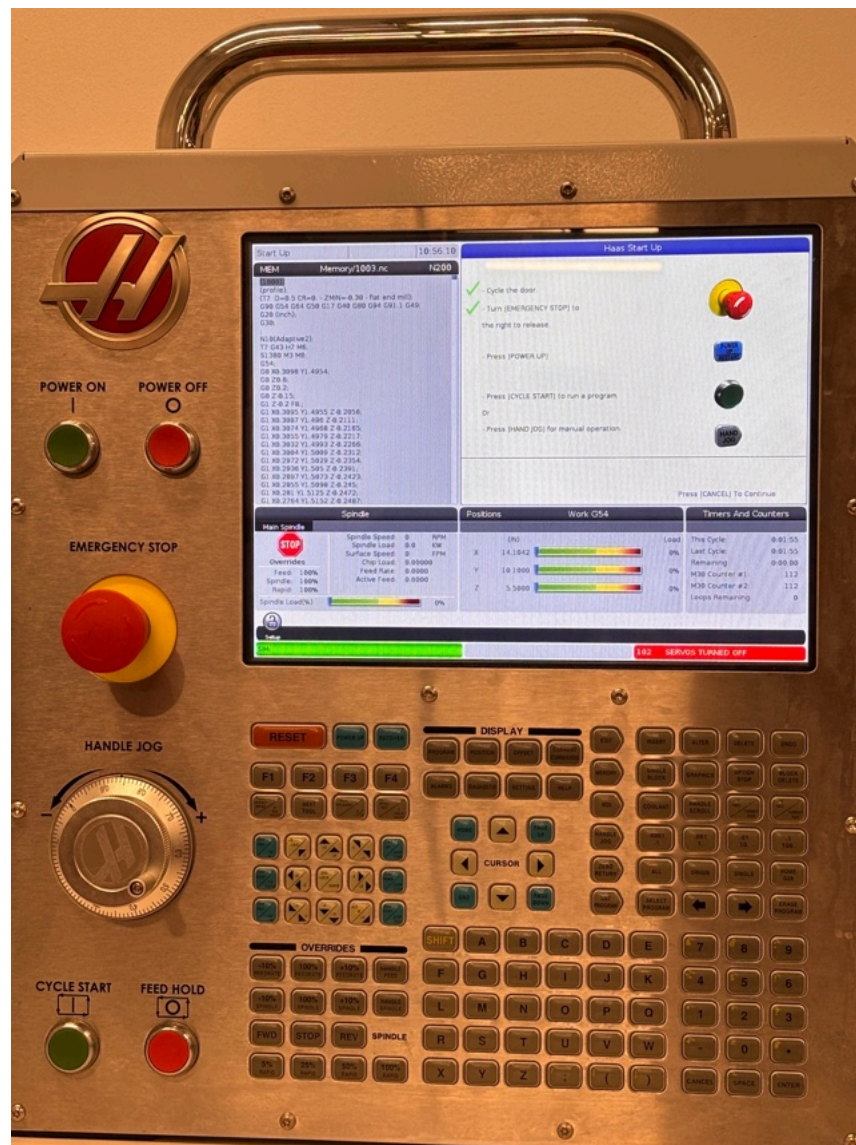
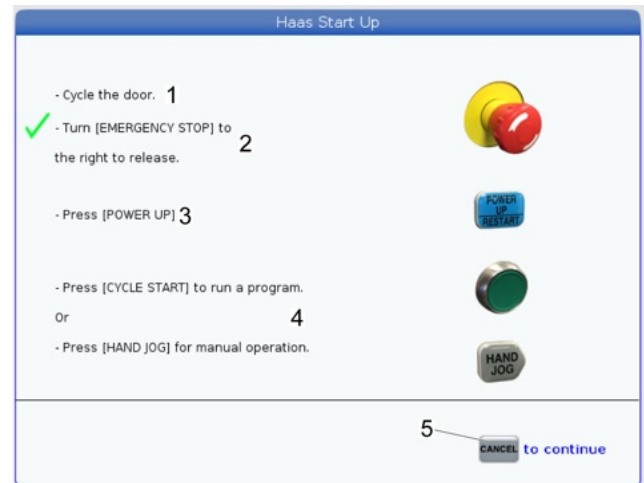


Table of Contents

Powering on the Machine	2
Machine Controls and Menu	3
Loading a Program	6
Graphics Verification	7
Tool Setup and Probing	7
Workpiece Setup and Probing	8
Coolant System	8
Preparing the Machine	9

Powering on the Machine

1. Press the **Power On** button.
2. Follow the startup instructions displayed on the side of the machine.
3. Verify that the **Emergency Stop (E-Stop)** is released.
4. Cycle the machine doors.
5. Press **Power Up**.
6. The machine will automatically home all axes and perform a tool change.
7. Once homing is complete, all machine menus are available.



Machine Controls and Menu

Handle Jog

Allows manual movement of the machine.

1. Select the desired axis using the axis keypad.
2. Choose the jog increment
3. Rotate the handwheel to move the selected axis.



Program

Displays the current program and machine information. Always review the program before machining to make sure its the right one.



Machine

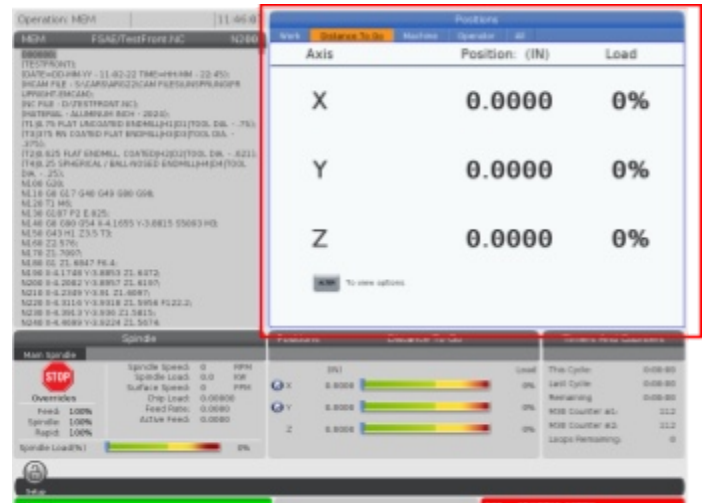
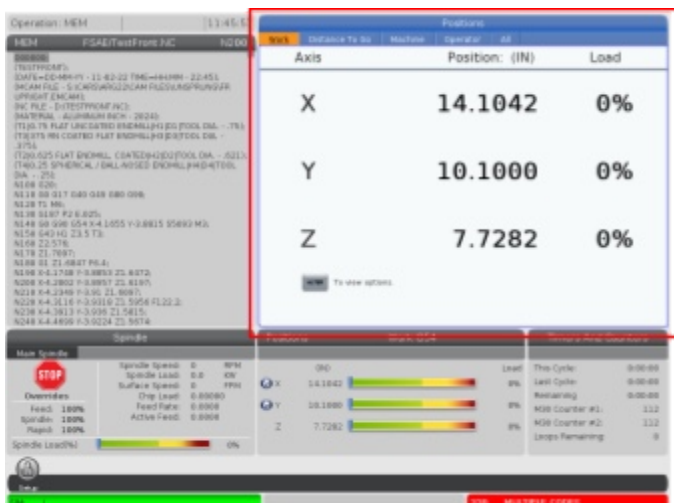
Displays machine positions.

Distance to Go

Shows the machine's next commanded position and the remaining travel distance.

Position Table

Displays the current X, Y, and Z position.



Offset

Contains Work offsets and Tool offsets

Work Offsets

Typically contain negative values defining the work coordinate system.

Tool Offsets

Stores measured tool lengths and diameters. Large or heavy tools may be designated as Large Tool or Heavy Tool, which allows additional clearance and slower tool changes. Tool speed limits may also be adjusted here.

Offsets

Tool Work

Active Tool: 1

Tool Offset	Length Geometry	H(Length) Wear	Diameter Geometry	Diameter Wear
1 Spindle	-2.3544	0.	0.7521	0.
2	-3.6400	0.	0.6227	0.
3	-6.3700	0.	0.7523	0.
4	-4.7400	0.	0.2514	0.
5	-3.5436	0.	0.2511	0.
6	-3.2589	0.	0.1960	0.
7	-6.3526	0.	0.7523	0.
8	0.	0.	0.	0.
9	0.	0.	0.	0.
10	0.	0.	0.	0.
11	0.	0.	0.	0.
12	0.	0.	0.	0.
13	0.	0.	0.	0.
14	0.	0.	0.	0.
15	0.	0.	0.	0.
16	0.	0.	0.	0.
17	0.	0.	0.	0.
18	0.	0.	0.	0.

Changes may not exceed setting 142 while machine is running.

TOOL
OFFSET
MEASURE

Tool Offset Measure

F1

Set Value

ENTER

Add To Value

F4

Work Offset

Offsets

ToolWork

Axes InfoWork Probe

G Code	X Axis	Y Axis	Z Axis		
G52	0.	0.	0.		
G54	-14.1042	-10.1000	-5.5000		
G55	-7.7784	-10.1000	-5.5000		
G56	0.	0.	0.		
G57	0.	0.	0.		
G58	0.	0.	0.		
G59	0.	0.	0.		
G154 P1	0.	0.	0.		
G154 P2	0.	0.	0.		
G154 P3	0.	0.	0.		
G154 P4	0.	0.	0.		
G154 P5	0.	0.	0.		
G154 P6	0.	0.	0.		
G154 P7	0.	0.	0.		
G154 P8	0.	0.	0.		
G154 P9	0.	0.	0.		
G154 P10	0.	0.	0.		
G154 P11	0.	0.	0.		
G154 P12	0.	0.	0.		

F1Set Value

ENTERAdd To Value

F4Tool Offset

Changes may not exceed setting 142 while machine is running.

Display

Provides active alarms, alarm history, key history, diagnostic information, lubrication status, QR codes for troubleshooting, and help information for machine alarms.

Settings

Used to configure machine settings.

List Program

Displays machining programs stored in machine memory, user data, and also in the USB (if plugged in)

Edit / VPS

Allows editing G-code and creating tool or work offsets. VPS stands for Visual Programming System.



Used for short commands and quick operations. For example, entering T5 followed by ATC Forward changes to Tool 5.

Overrides

Used to adjust Feedrate, Spindle Speed, and Rapid Speed. A common starting point is 25% Rapid Override. Adjust feedrate based on chip formation and cutting sound.

Additional Controls

Block Delete

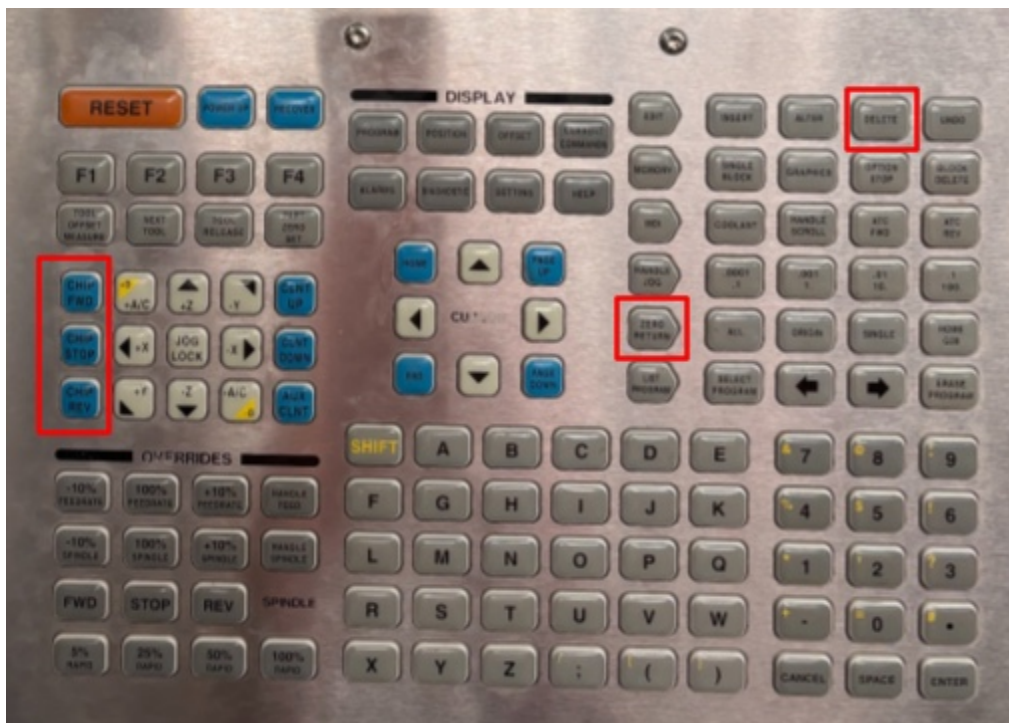
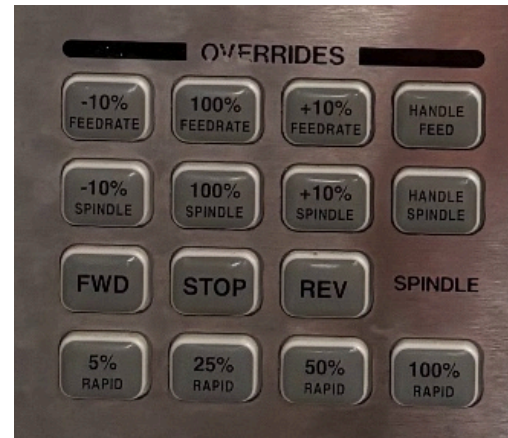
Deletes highlighted lines while editing.

Zero Return

Returns machine axes to home. To home a single axis, select the desired axis and press G28 Zero Return. If no axis is selected, all axes will return home.

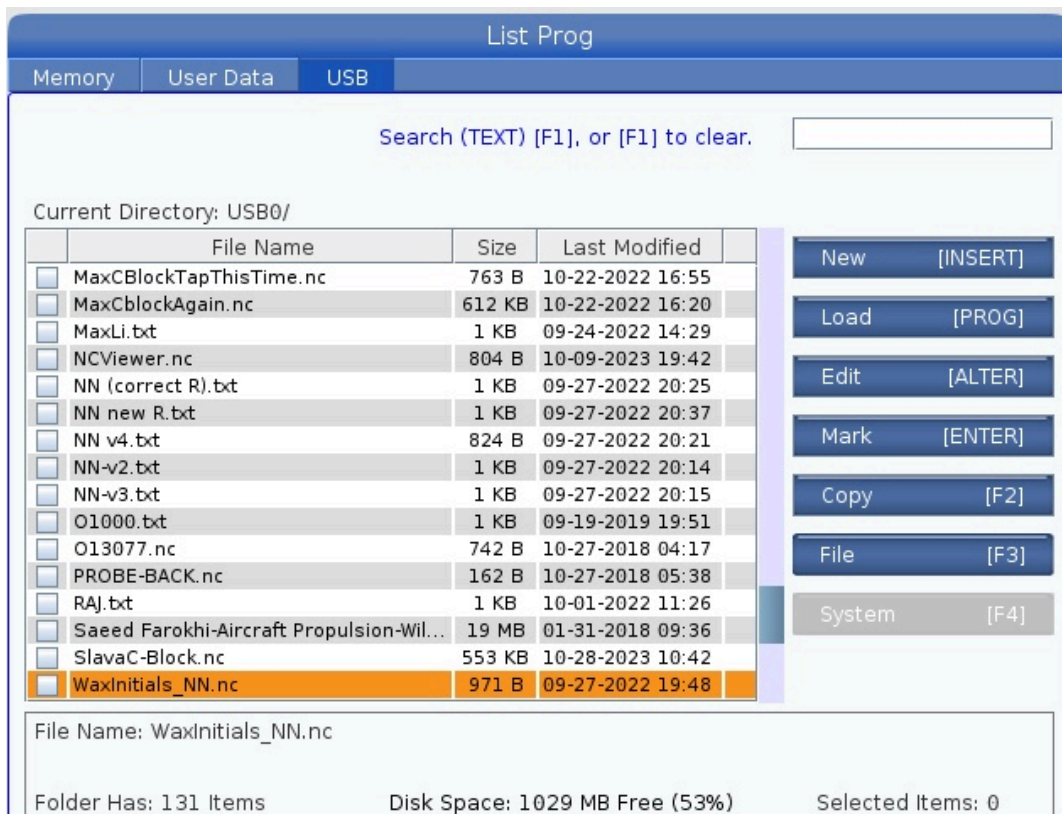
Chip Auger

Moves the chip auger forward or backward. Always operate the auger with coolant flowing to help clear chips. The machine door must be closed.

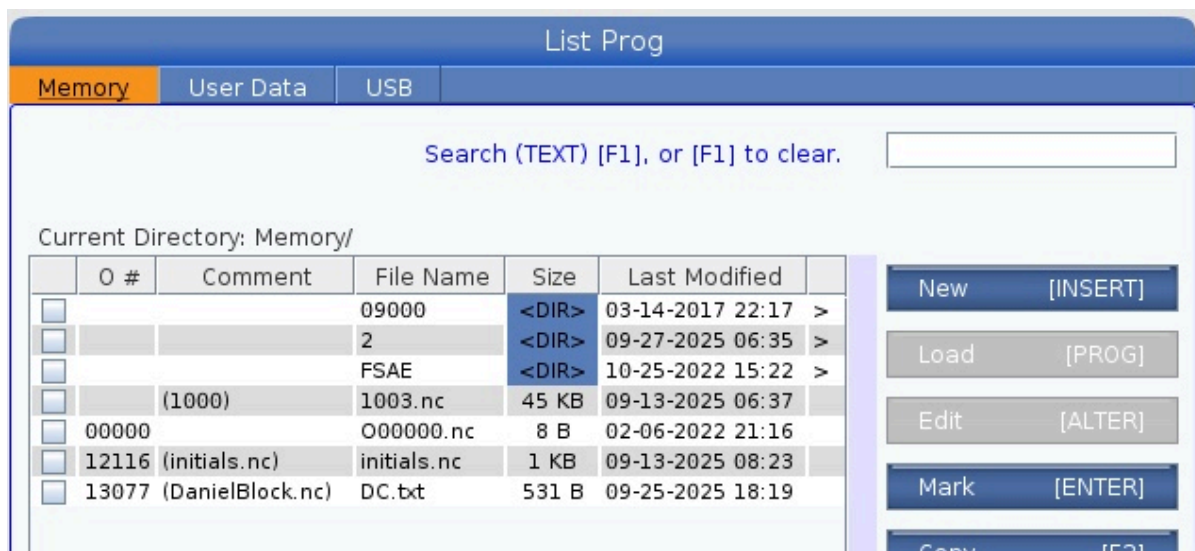


Loading a Program

1. Insert the USB drive.
2. Press **List Program** twice.
3. Navigate to the USB drive.
4. Locate the desired program using the arrow keys.
5. Press **F2** to copy the program.



6. Copy the file into **Memory**.



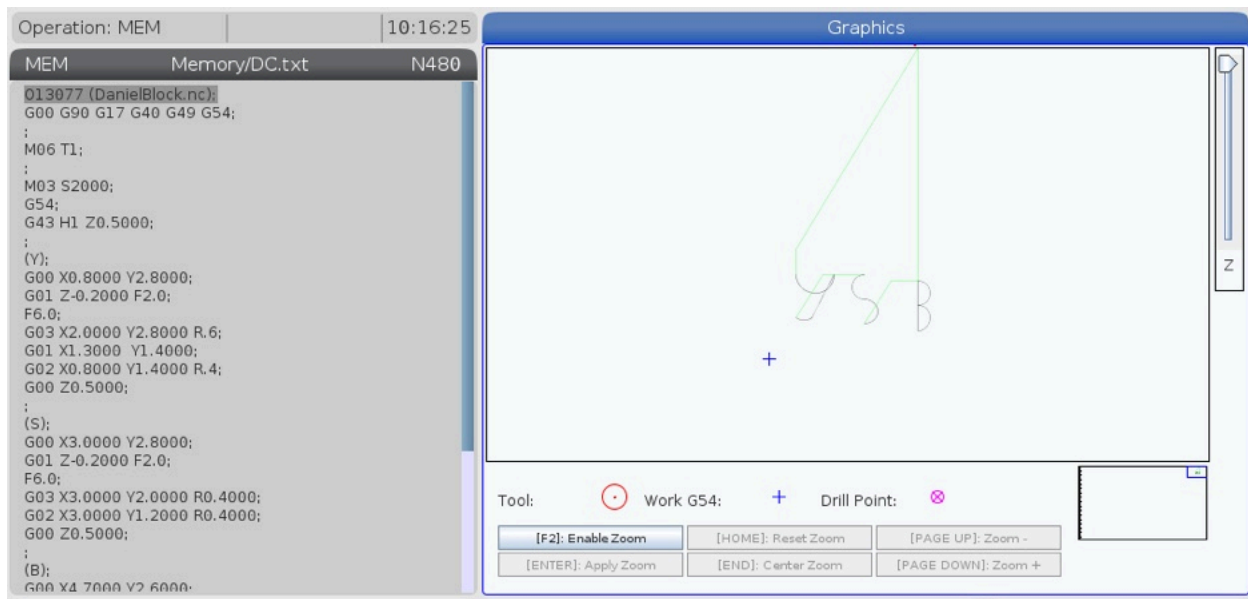
Notes

- **Memory** is the machine's internal storage.
- Never modify or open **Folder 09000**. It contains critical HAAS system programs.
- Do not store machining programs in **User Data**.
- Never run programs directly from a USB drive. Programs execute more reliably when stored in machine memory.

Graphics Verification

1. In Memory mode, open the Graphics display.
2. Press Reset to return to the beginning of the program.
3. Press Cycle Start to simulate the program.
4. Use the Graphics Speed keys to slow down or speed up the simulation as needed.

Graphics mode also checks the program for basic errors, such as missing feedrates or syntax mistakes, in addition to displaying the toolpath. A good practice is to run graphics in a single block to help debug errors in your program and see tools paths. Press the Single Block button while in graphics mode to toggle between this.



If modifications are required:

1. Press Edit.
2. Highlight the desired line(s).
3. Type the corrected code.
4. Press Alter to replace existing code, Enter to insert new lines, or Delete to remove lines.

Tool Setup and Probing

Installing Tools

Install cutting tools into the ER collets and tighten using a torque wrench set to around 120 ft-lb. You should use the collet holder as seen in the picture. After use, remember to return the torque wrench to 0 to preserve the internal spring.

Changing Tools on HAAS

1. Enter MDI mode.
2. Type T# (where # is the tool number).
3. Press ATC Forward or ATC Reverse.

Probing Tools

1. Change to the desired tool.
2. Open Probing, then Tool Offsets.
3. Select the appropriate probing cycle and follow the on-screen prompts.
4. Close the machine doors.
5. Press Cycle Start.



Program Generation

Editor VPS

VPS

To Switch Boxes [F4]

Load [ENTER]

Back Forward Search (TEXT) [F1], or [F1] to clear.

Current Directory: PROBING/TOOL SETTING/

File Name	Size	Last Modified
Probe Manual Length	2358	01/26/17 01:57
Auto Length, Non-rotating	7181	01/26/17 01:57
Auto Length, Rotating	12125	01/26/17 01:57
Auto Length and Diameter	12890	01/26/17 01:57
Auto Length, Sequential	2875	01/26/17 01:57
Auto Length, Random	11281	01/26/17 01:57
Tool Breakage Length, Non-rotating	3847	01/26/17 01:57

Workpiece Setup and Probing

Loading a Work Piece

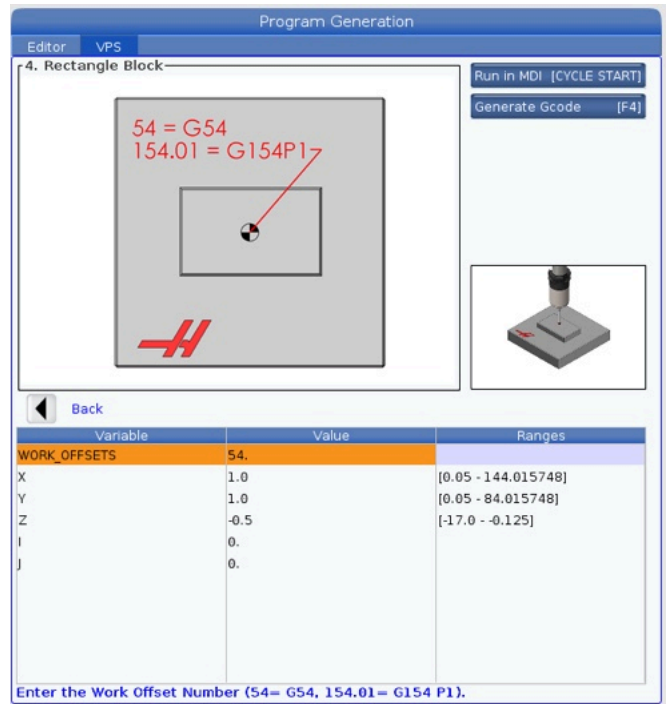
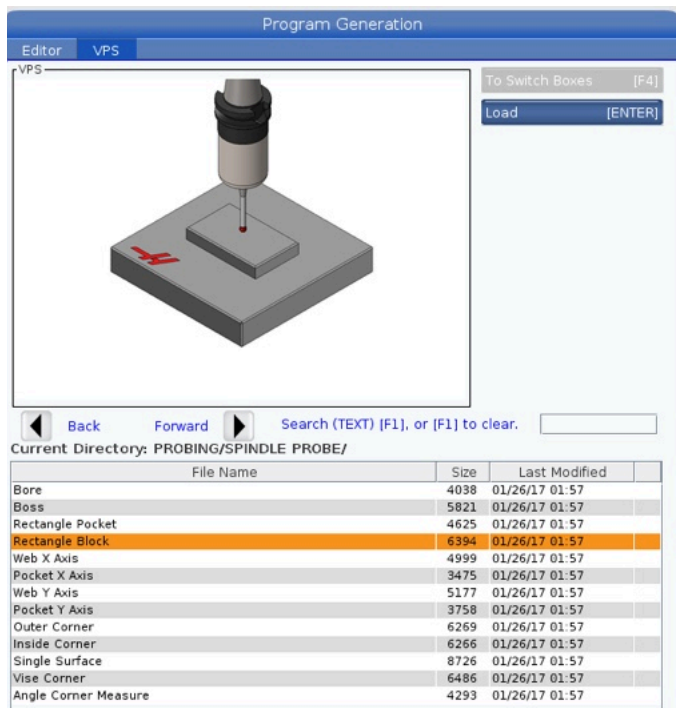
1. Install the spindle probe.
2. Use Handle Jog to position the vise beneath the probe.
3. Clamp the workpiece securely.

Example: Rectangular Block

1. Open Edit → Probing → Work Offsets → Rectangle Block.
2. Position the probe over the center of the workpiece.
3. Probe the Z surface. By selecting the single surface setting in the workoffsets menu.

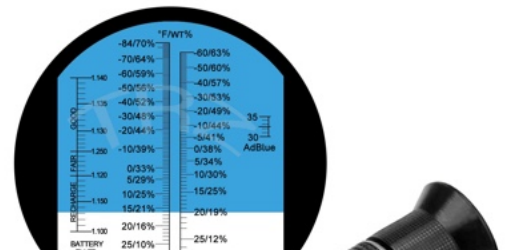
If holes prevent probing the top surface, use the Single Surface probing routine.

Important: Never start the spindle or coolant while the spindle probe is installed.



Coolant System

Coolant Nozzle



Use the Coolant button to toggle coolant flow. Adjust the coolant nozzle using Coolant Up or Coolant Down. Verify the required nozzle position using the coolant position column on the Offset page.

Measuring Coolant Concentration

1. Collect coolant with a pipette.
2. Place a sample on the refractometer.
3. Hold the refractometer toward a light source.
4. Read the boundary between the blue and white regions.

Recommended concentration is 8–15%.

Preparing the Machine

1. Secure the stock in the center of the vise.
2. Leave enough material below the part to create a sacrificial tab or "cleat" that can be removed after the flip-and-face operation.
3. Install the required tools into the appropriate tool pockets.
4. Probe all tools and set the work offset (see **Tool and Work Setup**).
5. Insert the USB drive containing your G-code.
6. Copy the program from the USB drive into the machine memory.
7. Open the program in **Graphics Mode** and verify the toolpath.
8. Edit the code if necessary.
9. Check the coolant concentration using a refractometer and dilute with water if required.
10. Run the program.
11. Adjust rapid, spindle, and feedrate overrides as needed to improve machining performance.
12. If desired, enable **Single Block** mode to execute one line of code at a time.
13. Ensure the chip auger is running whenever coolant is spraying.
14. Continuously monitor the machining operation.
15. If something appears incorrect, press **Feed Hold** immediately.

Warning: Never press the **Emergency Stop** while the tool is actively cutting unless absolutely necessary. Abruptly stopping the machine under load can damage tooling or the machine.